Implementing an Impedance Compression Network to Compensate for Misalignments in a Wireless Power Transfer System

Jungwon Choi, *Student Member, IEEE*, Jiale Xu, *Student Member, IEEE*, Rawad Makhoul, *Student Member, IEEE*, and Juan M. Rivas Davila, *Member, IEEE*

***Abstract*—This paper presents an impedance compression network (ICN) design in a wireless power transfer (WPT) system to compensate for distance or alignment variations between coils. In midrange WPT applications, magnetic resonant coupling coils are mainly implemented to achieve high efficiency for charging batteries. However, a distance or horizontal alignment variation between coils changes their coupling coefficient, and it decreases the overall performance of WPT systems because the zero voltage switching in a resonant inverter is lost. In order to reduce coil impedance variation, we propose an ICN that compresses variations in coil impedance. The ICN consists of a resistance compression network and a phase compression network to suppress magnitude and phase variations, respectively, in the coil impedance. Only passive components, such as inductors and capacitors, are used to implement an ICN.**

***Index Terms*—Impedance compression network, magnetic resonant coupling, resonant inverter, wireless power transfer.**

I. Introduction

RECENTLY, wireless power transfer (WPT) for automated guided vehicles (AGVs) has been drawing a lot of attention because it eliminates the need for manual intervention to charge AGV batteries [1], [2]. In order to embed a WPT system in an AGV, the overall volume and weight of the system must be small. Increasing the switching frequency to tens of MHz reduces weight and improves power density because the size of passive components, such as inductors and capacitors, are inversely proportional to the switching frequency.

However, fast switching operation increases switching losses in conventional hard-switching converters. In order to reduce switching losses, soft-switching inverters, such as a class E or class Φ_2 inverter [3]–[9], have been used to operate WPT systems at MHz switching frequency. In particular, [8] demonstrated a 1.3 kW class Φ_2 inverter operating at 13.56 MHz switching frequency with 94% efficiency. Also, [10] describes a 2-kW class Φ_2 inverter with a 900 V SiC MOSFET operating at 13.56 MHz switching frequency. Fig. 1(a) and (b) shows a single-ended class E and a single-ended class Φ_2 inverter, respectively. They are designed to achieve zero voltage switching (ZVS) and zero *dv/dt* (*Zdv/dt*) operation that decrease switching losses by eliminating overlapping voltage and current, which occur simultaneously in switching devices.

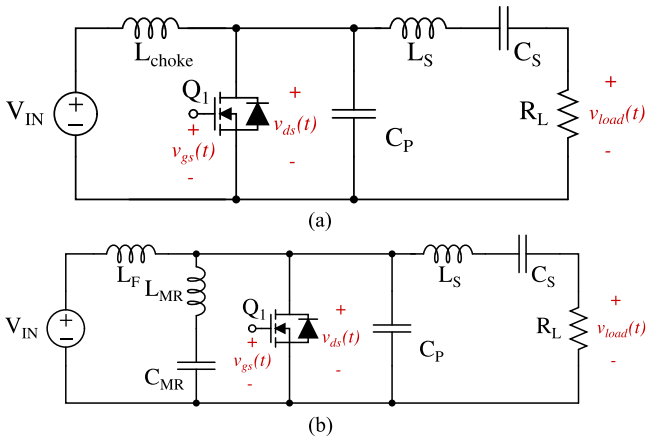


Fig. 1. Single-ended resonant inverters. (a) Class E inverter. (b) Class Φ_2 inverter.

But these inverters are sensitive to load variations, which may lower their efficiency as a load values or alignment between transmitters and receivers varies. In a WPT design, the coil impedance presents a load for an inverter and it is tuned to be resistive at a specific frequency with a high-quality ratio (a high Q system). Therefore, small variations in coil alignments (vertical and horizontal) can cause load variations that eliminate ZVS and zero *dv/dt* operation in a resonant inverter.

For example, magnetic resonant coupling (MRC) coils can achieve high efficiency above 90%, delivering power over a wide range of air gaps, even at a distance larger than the diameter of the coupling coils [11]–[16]. However, when they are connected to a resonant inverter and any misalignment between coils exists, the efficiency of the inverters may decrease because

Manuscript received October 6, 2017; revised February 7, 2018 and May 8, 2018; accepted July 24, 2018. Date of publication August 1, 2018; date of current version March 29, 2019. This work was supported by Daihen Corporation through the Stanford SystemX Alliance FMA (Faculty, Mentor, Advisor) Research Program. Recommended for publication by Associate Editor O. C. Onar. *(Corresponding author: Jungwon Choi.)*

J. Choi, J. Xu, and J. M. Rivas are with Stanford University, Stanford, CA 94305 USA (e-mail: jwonchoi@stanford.edu; jialexu@stanford.edu; jmrivas@stanford.edu).

R. Makhoul is with the CEA-The French Commission for Atomic and Alternative Energies and Grenoble Institute of Technology (Grenoble INP), Grenoble 38031, France (e-mail: Rawad.MAKHOUL@cea.fr).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TPEL.2018.2862414

0885-8993 © 2018 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

of the sensitivity due to high Q of the coils. Specifically, when the horizontal or vertical alignment between coils varies, the coil impedance is not a resistive load for inverters, but a capacitive or inductive load. Therefore, alignment variations between coils eliminate ZVS operation in resonant inverters causing additional losses.

In order to minimize the sensitivity of MRC coils, the following methods were proposed. First, variable capacitors or a capacitor matrix with switching devices was used in an MRC coil system [17]. Also, mutual inductance in a four-coil system was adjusted to tune the resonant frequency [18]. To maintain the high efficiency of a resonant inverter, frequency tracking was implemented with a phase shift function [19]. Even though these methods successfully reduced the MRC-coil sensitivity, a complicated tracking or controlling system was required, making it difficult to operate at MHz switching frequencies.

In this paper, we propose an impedance compression network (ICN) that narrows the range of the load variation due to distance or horizontal alignment deviations between MRC coils in a WPT system without a complicated control system. An ICN consists of a resistance compression network (RCN) and phase compression network (PCN) [20]–[22]. We developed the ICN concept by combining the RCN and PCN, and scaling up the power level to compress the impedance variations in WPT systems for EVs. While the RCN reduces the load variations in the MRC coils through a matching and transformation network, it is only validated when output loads are resistive. When any variations in the MRC coils occur, the load impedance for the inverter may have capacitive or inductive terms, causing phase shifts. To address the phase shift resulting from a distance or horizontal alignment variation, we designed and implemented a PCN using a Smith chart. Therefore, an ICN simultaneously provides magnitude and phase compression when the load impedance varies in MRC coils.

Section II describes the configuration and design methodology of the ICN with MRC coils. Section III presents a preliminary MRC coil design and its simulated results in Fasthenry and LTSPICE. Section IV shows experimental results of an ICN with MRC coils. The simulated results of the class Φ_2 inverter connected to the ICN and MRC coils are described in Section V. Section VI shows the experimental results of an ICN with a class Φ_2 inverter. Section VII concludes this paper.

II. Managing Load Impedance With ICN in MRC coils

The advent of MRC allows us to increase the distance between two coils in a WPT system while maintaining high efficiency. However, existing MRC-WPT systems have not been efficient enough to power EVs, AGVs, and other large-scale applications when any variations of more than 5% between transmitting and receiving coils occur. In other words, even though an MRC-WPT system can provide high efficiency, variations in distance or horizontal alignment change the coil impedance $Z_{\rm IN}$ and significantly degrade the performance of a power source such as a resonant inverter.

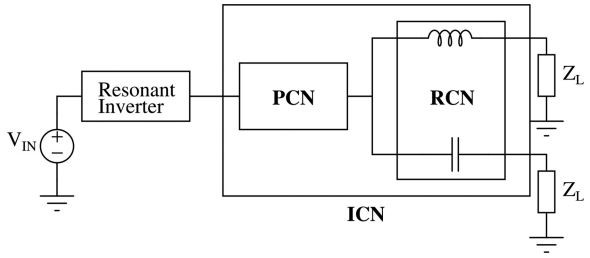


Fig. 2. ICN configuration consisting of RCN and PCN.

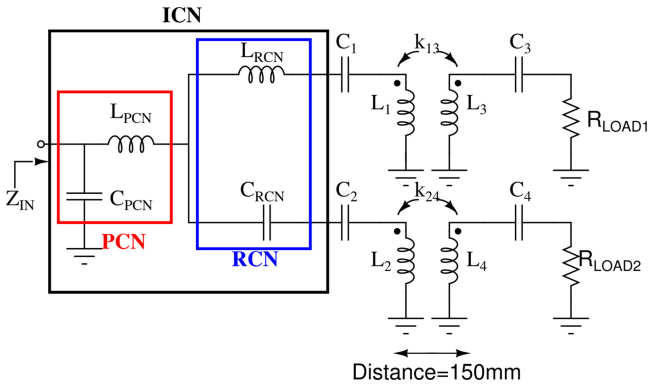


Fig. 3. Preliminary design of an ICN when two sets of MRC coils are used.

To address this issue in the MRC coils, we propose an ICN in a WPT system to minimize the effect on impedance when any distance or horizontal alignment variation occurs. The ICN configuration is shown in Fig. 2. When the distance or horizontal alignment between the MRC coils varies, the magnitude and phase of the coil impedance change simultaneously. The ICN compresses the magnitude and phase variations in two steps. First, based on the measured coil impedance with position deviations, the RCN compresses the magnitude variations of the coil impedance. Then the PCN compresses the phase variation. Another advantage of the ICN is that it consists of only passive components, such as inductors and capacitors that are ideally lossless components. Therefore, we can effectively mitigate the load variation in MRC coils without any complicated control systems and additional losses compared to the earlier approaches mentioned in Section I.

Fig. 3 shows the preliminary design of an ICN with MRC coils. L_1, L_2, L_3, and L_4 are the coils for WPT operation, and C_1, C_2, C_3 and C_4 are the capacitors to provide resonance at 13.56 MHz. $R_{\rm load1}$ and $R_{\rm load2}$ are 50 Ω resistive loads, and $Z_{\rm IN}$ is the impedance seen by a resonant inverter. To reduce the variation in $Z_{\rm IN}$, $L_{\rm RCN}$ and $C_{\rm RCN}$ must be chosen to compress the magnitude of the coil impedance, and $L_{\rm PCN}$ and $C_{\rm PCN}$ in PCN must be selected to reduce phase shifts.

A. Compressing Magnitude Variations in MRC Coils

An RCN consists of inductors and capacitors, and ideally transfers energy to the resistive loads without any losses. Previous work [20] assumed that two loads have the same value and both are purely resistive. When looking at the load through an RCN, the range of the load value appears compressed. While

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

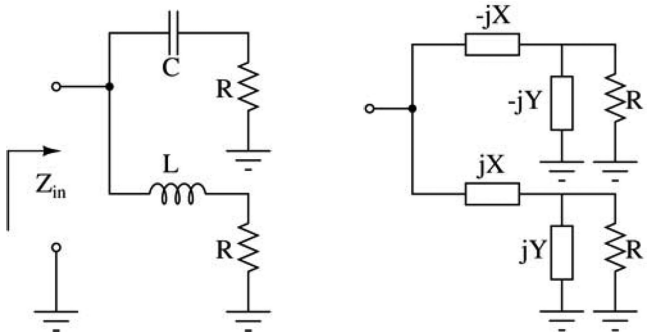


Fig. 4. Resistance compression topologies. (a) RCN with resistive loads. (b) Four-element compression network consisting of four passive components.

the loads R vary over a wide range, $Z_{\rm IN}$ presents a small amount of variation at resonant frequency w_o.

The impedance varies based on the following equations in Fig. 4(a):

$$Z_{\rm in} = \frac{Z_o^2}{2R}\left[1+\left(\frac{R}{Z_o}\right)^2\right] \tag{1}$$

$$w_o = \frac{1}{LC} \tag{2}$$

$$Z_o = \sqrt{\frac{L}{C}} = R \tag{3}$$

where Z_o is the characteristic impedance of the inductor and capacitor in an RCN [20]. For convenience, we set Z_o to the same as the load value as shown in (3). When *R* varies from Z_o/c to cZ_o (c is constant, defining the range of the load variation), we can rewrite $Z_{\rm IN}$ as

$$Z_{\rm in} = \frac{Z_o}{2}\left(c+\frac{1}{c}\right) \tag{4}$$

which provides a smaller variation in $Z_{\rm IN}$ than in R. We are able to not only compress the load variation, but also increase or decrease the impedance with a four-element network as shown in Fig. 4(b).

Two pairs of MRC coils become the two loads required by the RCN. One pair consists of two resonant capacitors (C_1 and C_3) and two coupling coils (L_1 and L_3). Another pair also consists of two resonant capacitors (C_2 and C_4) and two coupling coils (L_2 and L_4). Two resistance loads ($R_{\rm LOAD1}$ and $R_{\rm LOAD2}$) in the proposed system are actually the equivalent resistance of the rectifier connected to the battery in EVs. If there is only one resistive load in an EV, then two receiving coils are placed in series and connected to one rectifier and one battery. If there are several battery cells in EVs, then we can charge each battery cell separately.

Each pair is designed to provide a resonance at 13.56 MHz switching frequency when placed at 150 mm apart. We assume the following:

1) Two pairs of MRC coils are placed in parallel.
2) Coupling between two coils placed next to each other (i.e., k_{12} and k_{34}) is zero.
3) All inductance values and sizes of coils are the same.

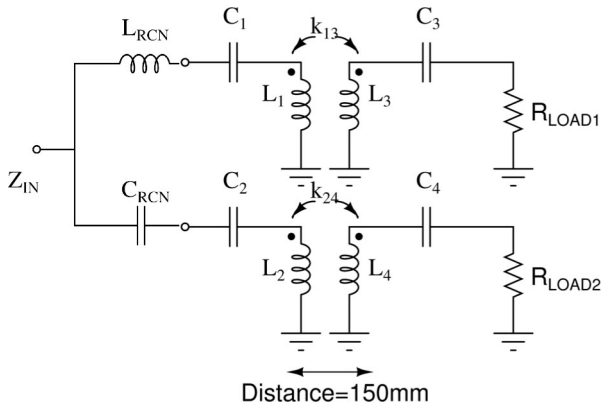


Fig. 5. MRC coils with an RCN.

Based on these assumptions, the impedance values are the same as the Z_o value used to design an RCN.

Fig. 5 shows the equivalent circuit of MRC coils with an RCN. When the distance or horizontal alignment between coils varies, the coupling coefficient also changes. For example, as the distance or horizontal misalignment increases, the coupling coefficient decreases, resulting in impedance variation. Then, the RCN in Fig. 5 effectively compresses the variation in the magnitude of the impedance. In the following equations, we replace the coil impedances (Z_1 and Z_2) with their complex expressions (R_1+jX_1 and R_2+jX_2) to calculate the impedance when using an RCN

$$\begin{aligned} Z_1 &= R_1 + jX_1 \\ Z_2 &= R_2 + jX_2. \end{aligned} \tag{5}$$

The input impedance ($Z_{\rm IN}$) is

$$Z_{\rm IN} = Z_{1,\rm rcn} || Z_{2,\rm rcn} \tag{6}$$

where

$$\begin{aligned} Z_{1,\rm rcn} &= jwL_{\rm rcn} + Z_1 \\ Z_{2,\rm rcn} &= \frac{1}{jwC_{\rm rcn}} + Z_2 \end{aligned} \tag{7}$$

and

$$\begin{aligned} L_{\rm rcn} &= \frac{Z_o}{w_s} \\ C_{\rm rcn} &= \frac{1}{Z_o w_s}. \end{aligned} \tag{8}$$

We assume that Z_1 is the same as Z_2

$$Z = Z_1 = Z_2 = R + jX. \tag{9}$$

Here is the calculation for the input impedance where $|Z|$ is the magnitude of Z

$$Z_{\rm IN} = R_{\rm IN} + jX_{\rm IN} \tag{10}$$

$$R_{\rm IN} = \frac{R(R^2+X^2+Z_o^2)}{2(R^2+X^2)} = \frac{R(|Z|^2+Z_o^2)}{2|Z|^2} \tag{11}$$

$$X_{\rm IN} = \frac{X(R^2+X^2-Z_o^2)}{2(R^2+X^2)} = \frac{X(|Z|^2-Z_o^2)}{2|Z|^2}. \tag{12}$$

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

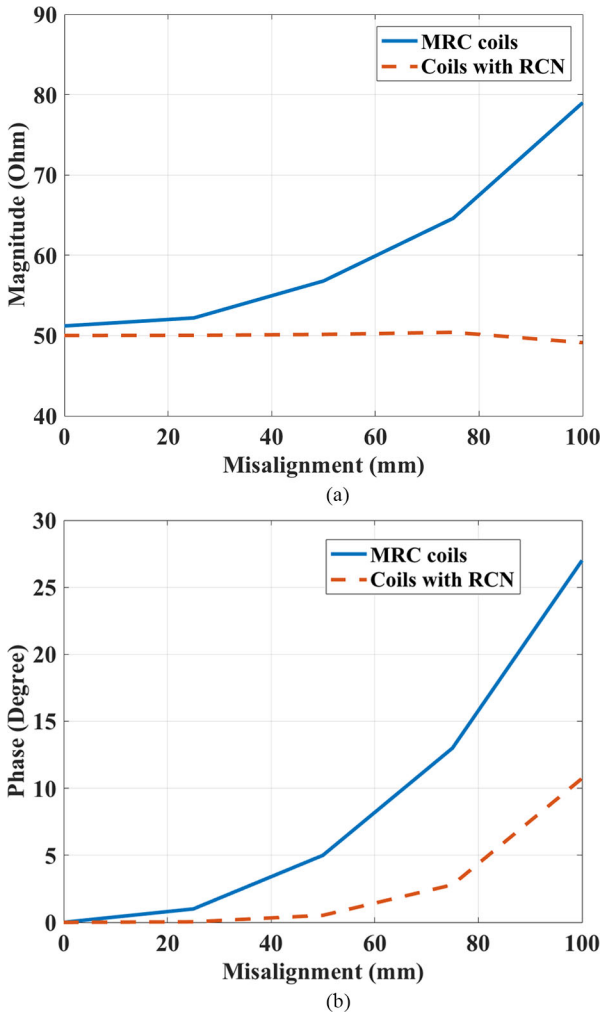


Fig. 6. (a) Calculated magnitude of Z_{in} with horizontal alignment variations. (b) Calculated phase of Z_{in} with horizontal alignment variations.

We can rewrite the resistance and reactance as a magnitude and phase

$$|Z_{\rm IN}| = \frac{\sqrt{(R^2+(X-Z_o)^2)(R^2+(X+Z_o)^2)}}{2\sqrt{R^2+X^2}} = \frac{\sqrt{(|Z_L|^2+Z_o^2-2Z_oX)(|Z_L|^2+Z_o^2+2Z_oX)}}{2|Z|} \tag{13}$$

$$\text{Phase} = \tan^{-1}\left(\frac{X(R^2+X^2-Z_o^2)}{R(R^2+X^2+Z_o^2)}\right) = \tan^{-1}\left(\frac{X(|Z|^2-Z_o^2)}{R(|Z|^2+Z_o^2)}\right). \tag{14}$$

Fig. 6(a) and (b) shows the calculated magnitude and phase of the input impedance when the horizontal alignment between coils varies. The measured coil impedances were used for Z_1 and Z_2. The solid lines show the measured coil impedance without the RCN and the dash lines show the calculated coil impedance when the RCN is added. As shown in these figures, the magnitude of the load impedance is effectively compressed with different alignments. However, we still have a phase shift, although it is lower than the phase shift in the load impedance without an RCN.

The RCN can also compress the load variations if two load values are not identical. Assuming that two loads are purely resistive, and R_1 is R and R_2 is $R+\Delta R$, then the impedance Z is

$$Z = (jwL_{\rm RCN}+R)//\left(\frac{1}{jwC_{\rm RCN}}+R+\Delta R\right) \tag{15}$$

$$= \frac{Z_o^2+R^2+\Delta R(jwL_{\rm RCN}+R)}{2R+\Delta R}. \tag{16}$$

For example, the ICN is designed for the parameters where Z_o and R are 50 Ω, w is 13.56 MHz, $L_{\rm RCN}$ is 587 nH, and $C_{\rm RCN}$ is 235 pF. If R varies from $0.5Z_o$ to $2Z_o$ and ΔR is zero, Z changes from Z_o to $1.25Z_o$. If ΔR is $0.1R$ (10% variation) and R varies from $0.5Z_o$ to Z_o, the magnitude of Z varies from Z_o to $1.18Z_o$. If ΔR is $0.1R$ (10% variation) and R varies from Z_o to $2Z_o$, the magnitude of Z varies from Z_o to $1.27Z_o$. This result indicates that the RCN can compress the magnitude variation when the load values are different from each other. Even though the load difference causes a phase shift in the impedance, the phase of the impedance varies by only a few degrees.

If we have harsh condition, such as one load value is five times larger than another load value, the load variation with the RCN is wider (7.5–17.5 Ω) than when we have two loads with the same value (10.5–12.5 Ω). However, the RCN can still compress load variations. Based on our experiments and analysis, two load values do not typically have values that differ widely because two receiving coils are placed on the same vehicle and they move in the same trend. In other words, the misalignments between L_1 and L_3, and between L_2 and L_4 are almost the same, so the difference in the two load values is not large.

B. Reducing Phase Shift in MRC Coils

As shown in the previous section, the RCN successfully reduces in the resistive load variation for the resonant inverter. However, it cannot compress the phase shift. In order to overcome this limitation, a PCN consisting of serial or parallel passive components was designed and added to the existing MRC coils and RCN. When a distance or horizontal alignment variation occurs, we can measure the impedance variations and draw the load line on a Smith chart. For example, if we change the distance between coils from 100 mm to 200 mm by 20 mm interval, six points can be marked in the Smith chart. When connecting all points, we can draw the load line, as shown in Fig. 7(a). As shown in Fig. 7(a), we can rotate the load line to the real axis on the Smith chart by adding the passive components in series or parallel to the MRC coils with the RCN. In other words, using one PCN allows us to rotate all points to the real axis. The red-solid line shows the measured load impedance without a PCN when the horizontal alignment between the transmitting and receiving coils varies. By adding the PCN (one pair of an *LC* network) as shown in Fig. 7(b), the red line can be rotated to the blue-dash line and eventually aligned with the real axis on the Smith chart.

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

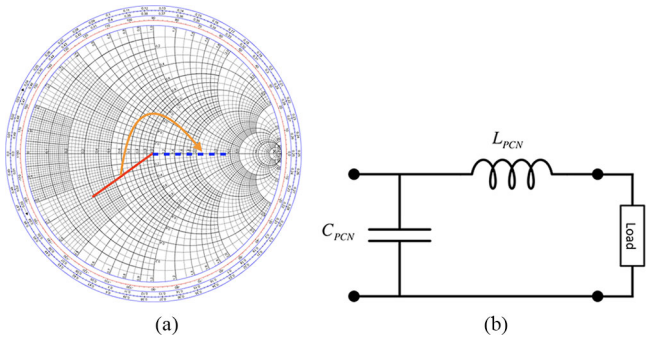


Fig. 7. (a) Original load line (red-solid line) and the rotated load line (blue-dash line) on the Smith chart. (b) PCN consisting of one inductor and one capacitor.

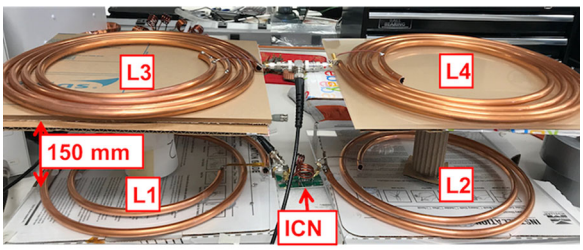


Fig. 8. Implemented MRC coils with the ICN.

Based on the measured impedance, either an inductor in series or a capacitor in parallel can be connected to the MRC coils. For example, if the load line has capacitive terms as in Fig. 7(a), we can add the serial inductor first. Otherwise, the capacitor can be connected first in parallel to the load and then we can place the inductor in series. Depending on the load line, an N-stage *LC* network can be included.

Here, we assume the load shown in Fig. 7(b) is capacitive

$$R_{\mathrm{Load}_i} = R_{L,i} - jX_{L,i}, 1 < i < n \tag{17}$$

where n is the number of measured load value with different distances or horizontal alignments. Then, the load impedance seen by the inverter is

$$Z_{\mathrm{PCN},i} = R_{\mathrm{PCN},i} + jX_{\mathrm{PCN},i} \tag{18}$$

$$= \frac{1}{jwC_{\mathrm{PCN}}} // [jwL_{\mathrm{PCN}} + (R_{L,i} - jX_{L,i})]. \tag{19}$$

The condition to compress the impedance variation is

$$\frac{R_{\mathrm{PCN},i}}{R_o} \approx 1, X_{\mathrm{PCN},i} \approx 0 \tag{20}$$

where R_o is the reference impedance value for designing the inverter. L_{PCN} and C_{PCN} are selected to fulfill these requirements.

By only adding passive components to MRC coils, an ICN can effectively compress impedance variations when the distance or alignment between transmitting and receiving coils causes changes in load impedance. The compressed impedance values allow us to reduce losses in a resonant inverter. Also, a complicated tracking system is not required in an MRC-WPT system. The ICN can be optimized by combining passive components to reduce the number of components.

TABLE I
COUPLING COEFFICIENT CALCULATED IN FASTHENRY WITH DIFFERENT DISTANCES BETWEEN A TRANSMITTING AND RECEIVING COIL

Distance (mm)	Coupling coefficient
50	0.433
75	0.309
100	0.228
125	0.172
150	**0.132**
175	0.103
200	0.081

TABLE II
COUPLING COEFFICIENT CALCULATED IN FASTHENRY WITH DIFFERENT HORIZONTAL ALIGNMENTS BETWEEN A TRANSMITTING AND RECEIVING COIL

Horizontal alignment (mm)	Coupling coefficient
0	**0.132**
25	0.129
50	0.121
75	0.11
100	0.097
125	0.081

III. PRELIMINARY DESIGN OF MRC COILS

As shown in Fig. 3, two sets of MRC coils were simulated and implemented as a prototype design to operate at 13.56 MHz. First, the coupling coils were simulated in Fasthenry [23] to obtain inductance values of the coils and coupling coefficient between them. We selected coil specifications such as a diameter, a number of turns, a cross section area of the coils, and the distance between coils. Our target distance between transmitting and receiving coils was 150 mm and the diameter of the coil was 250 mm. The desired number of turns was two for the transmitting coils and four for the receiving coils. Based on these specifications, we wrote a script to generate their model in MATLAB and ran it in Fasthenry. Table I shows the calculated coupling coefficients k_{12} with different distances between a transmitting and receiving coil, and Table II shows the k_{12} when horizontal alignment between coils varies. As shown in these tables, as distance or horizontal misalignment increases, the coupling coefficient decreases. The calculated inductance of the transmitting coil was 1.68 μH and the calculated inductance of the receiving coil was 5.56 μH. The inductance value of each coil in Fasthenry was the same as the measured value obtained through an impedance analyzer.

Then, we simulated the MRC coils using the inductance values and coupling coefficients from Fasthenry in LTSPICE to obtain the coil impedance. An ac analysis was used to calculate the coil impedance while changing the coupling coefficients based on the values shown in Tables I and II. Two sets of MRC coils were required for the RCN operation, and each coil set was

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

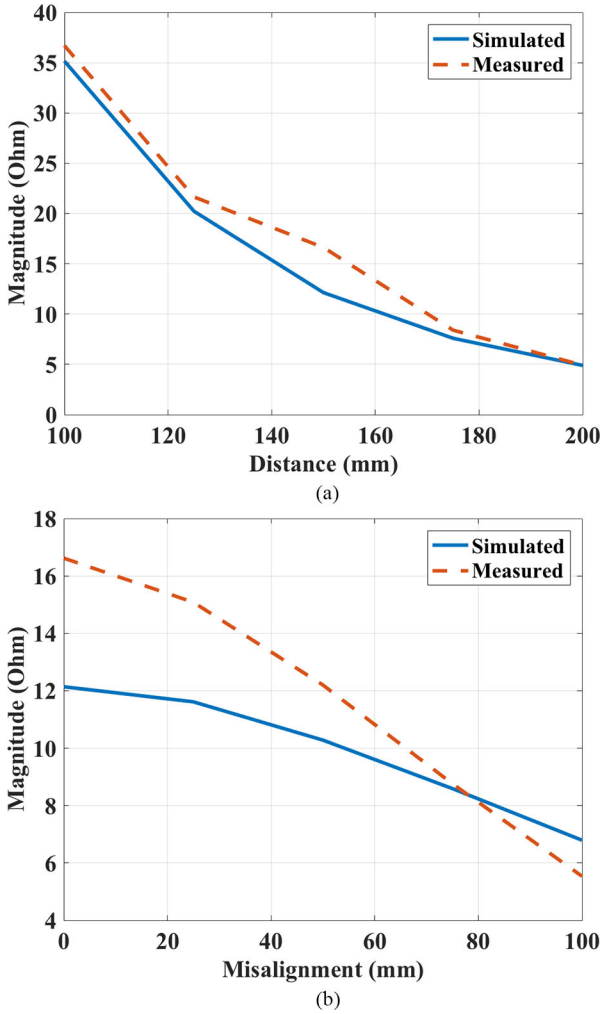


Fig. 9. Comparisons between simulated and measured coil impedance values. (a) Distance variations. (b) Horizontal alignment variations.

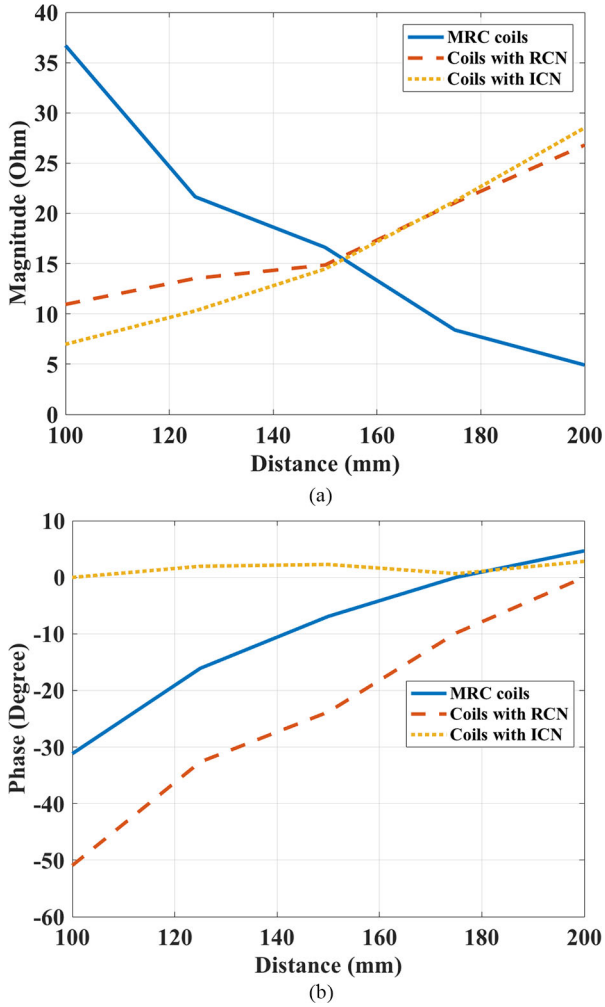


Fig. 10. Comparisons of the measured impedance values between MRC coils without the ICN, with the RCN, and with the ICN. (a) Magnitude and (b) phase with distance variations.

designed to be resonant at a switching frequency. The impedance of first coil set consisting of L_1, C_1, L_3, and C_3 was simulated with a 50 Ω load. The impedance of second coil set consisting of L_1, C_1, L_3, and C_3 was also simulated with a 50 Ω load. We selected 83 pF capacitors (C_1 and C_3) for transmitting coils and 25 pF capacitors (C_2 and C_4) for receiving coils for providing a resonance at 13.56 MHz. Fig. 9(a) and (b) shows the simulated impedance values of the MRC coils with distance and horizontal alignment variations in LTSPICE, respectively.

Finally, the MRC coils were implemented as shown in Fig. 8. Copper tubing was used for the coupling coils. In order to measure the coil impedance, we used the ENA series network analyzer, model E5061. For designing and tuning the coils, the impedance of each coil unit were separately measured. With a 50 Ω resistive load, measured Z_1 was 16 Ω and measured Z_2 was 15 Ω. The parasitic components and slightly different coil formation caused the difference between Z_1 and Z_2. Therefore, we set Z_o as 16 Ω, the average value of Z_1 and Z_2.

To characterize distance variation, we changed the distance between coils from 100 to 200 mm. For the horizontal misalignment experiments, we varied the horizontal alignment from 0 to 100 mm. Fig. 9(a) and (b) shows the comparisons between the simulated and in variations in MRC coil impedance with the distance and horizontal alignment changes, respectively. The measured MRC coil impedance varies in the range: $0.295 * Z_{\text{center}} < Z_{\text{IN}} < 2.21 * Z_{\text{center}}$, where Z_{center} is 16 Ω. To be specific, when the distance between the coils varies without the ICN, the impedance ratios are

$$\frac{|Z_{\text{IN}}|@200\text{ mm}}{|Z_{\text{IN}}|@150\text{ mm}} = 0.29, \frac{|Z_{\text{IN}}|@100\text{ mm}}{|Z_{\text{IN}}|@150\text{ mm}} = 2.2.$$

When the horizontal alignment between the coils varies without the ICN, the impedance ratios are

$$\frac{|Z_{\text{IN}}|@100\text{ mm}}{|Z_{\text{IN}}|@0\text{ mm}} = 0.3.$$

IV. Experimental Results of MRC Coils With an ICN

In this experiments, the reference magnitude of the MRC coil impedance for designing an ICN was 16 Ω. This value was measured when the distance between coils was 150 mm and the frequency was 13.56 MHz. Based on the reference value, we set

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

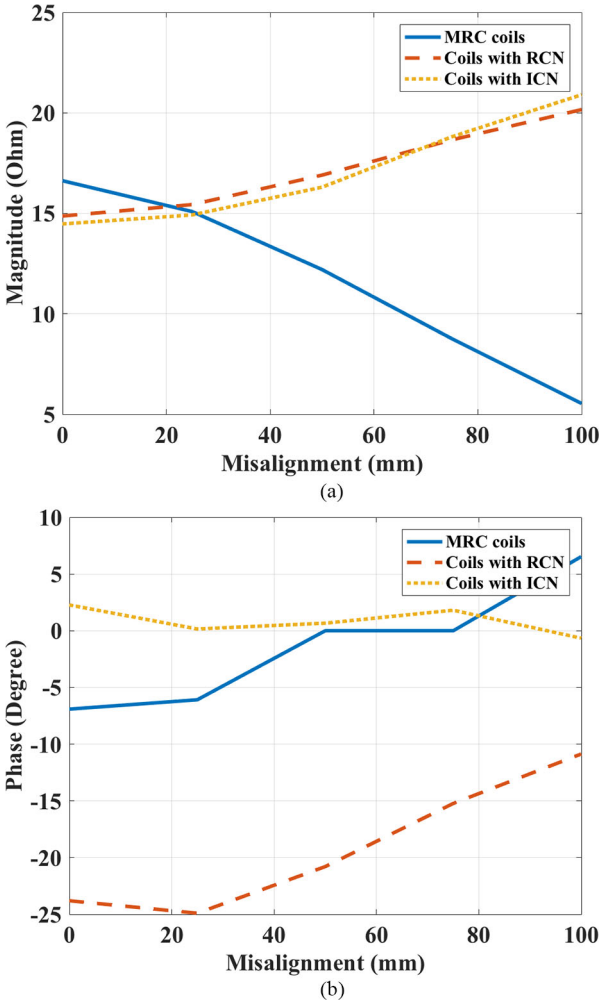


Fig. 11. Comparisons of the measured impedance values between MRC coils without the ICN, with the RCN, and with the ICN. (a) Magnitude and (b) phase with horizontal alignment variations.

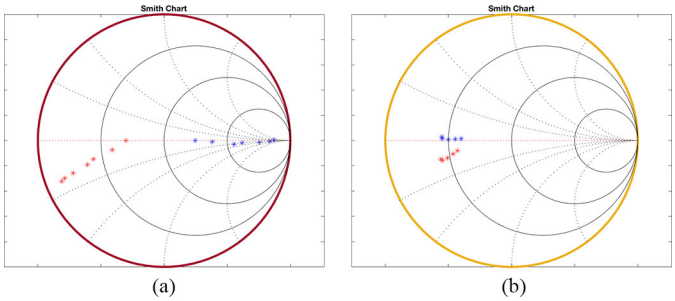


Fig. 12. Impedance variation after connecting the PCN. (a) Distance variations. (b) Horizontal alignment variations.

Z_o to 16 Ω and w_o to 13.56 MHz for designing the RCN. L_{RCN} (188 nH) and C_{RCN} (733 pF) were connected to the MRC coils as shown in Fig. 3. Then, we measured the impedance values of the MRC coils with the RCN when the distance or horizontal alignment between coils varied, and compared them to the values without the RCN.

Fig. 10(a) and (b) shows the magnitude and phase variations with different distances between coils with the added RCN.

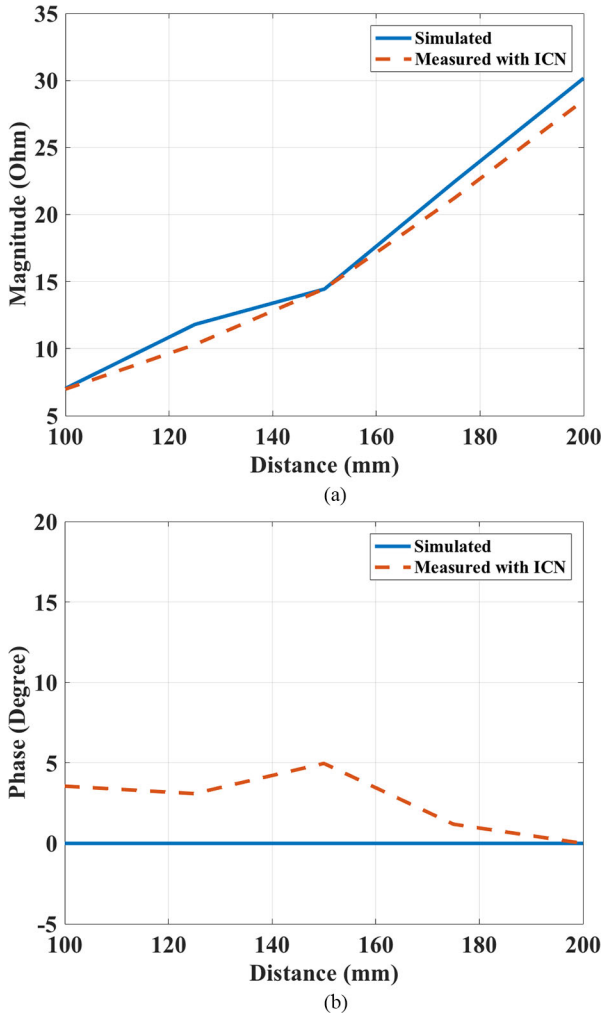


Fig. 13. Comparisons between simulated and measured coil impedance values. (a) Magnitude and (b) phase with distance variations.

Also, Fig. 11(a) and (b) shows the impedance variations with different horizontal alignment between coils with the added RCN. While the magnitude of the coil impedance without the RCN decreased with increasing in the distance or horizontal misalignment, the magnitude of the impedance with the distance or horizontal misalignment increased.

With the RCN, the MRC coil impedance varies in the range: $0.74 * Z_{\mathrm{center}} < Z_{\mathrm{IN}} < 1.8 * Z_{\mathrm{center}}$, where Z_{center} was 29 Ω. When the distance between the coils changes with the RCN, the impedance ratios are

$$\frac{|Z_{\mathrm{IN}}|@200\text{ mm}}{|Z_{\mathrm{IN}}|@150\text{ mm}} = 1.8, \frac{|Z_{\mathrm{IN}}|@100\text{ mm}}{|Z_{\mathrm{IN}}|@150\text{ mm}} = 0.74.$$

When the horizontal alignment between the coils changes with the RCN, the impedance ratios are

$$\frac{|Z_{\mathrm{IN}}|@100\text{ mm}}{|Z_{\mathrm{IN}}|@0\text{ mm}} = 0.794.$$

While the magnitude of the impedance effectively compressed, the phase shift in the coil impedance still existed.

In order to compress the phase shifts, we marked the measured impedance values using Smith chart, as shown in Fig. 12(a)

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

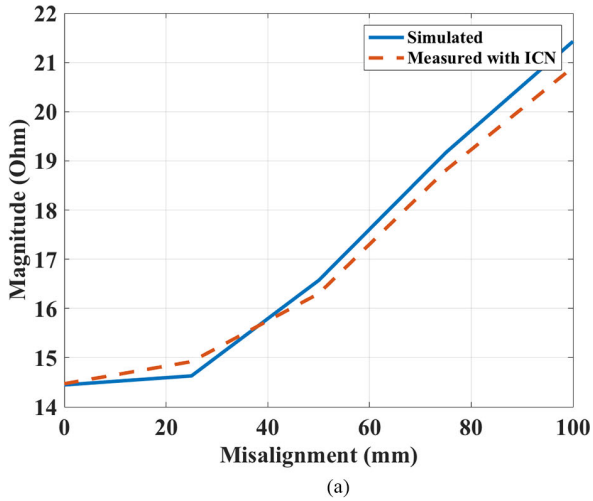


Fig. 14. Comparisons between simulated and measured coil impedance values. (a) Magnitude and (b) phase with horizontal alignment variations.

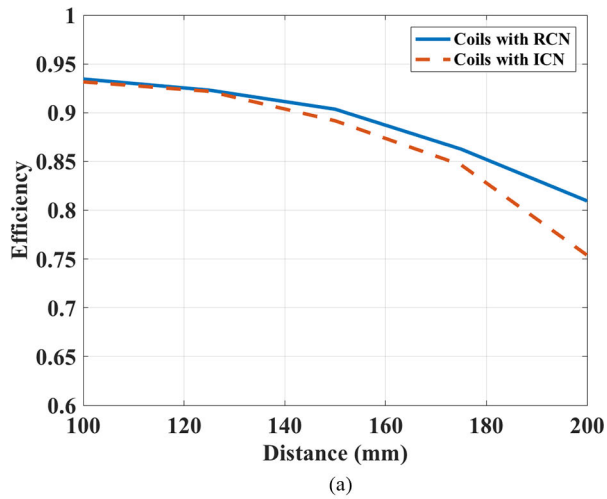


Fig. 15. Transmitting efficiency. (a) Distance variations. (b) Alignment variations.

and (b). The red dots show the impedance values before connecting a PCN. Because the impedance line looked capacitive, we first connected a 111 nH inductor in series with the RCN. Then a 138 pF capacitor was connected to the inductor and the RCN in parallel as shown in Fig. 3. These two components allowed us to rotate the impedance values, which were shown as red dots in Fig. 12(a) and (b), to the blue dots for aligning the load line with the real axis.

Figs. 10(b) and 11(b) show the phase variations with the ICN. With the ICN, we can compress not only the magnitude of the coil impedance, but also the phase of the impedance. Furthermore, we compared the measured impedance values with the ICN to the simulated values as shown in Figs. 13(a) and (b) and 14(a) and (b). The measured impedance values were matched with the simulated impedances well.

Finally, we measured scattering parameters (S parameters) to calculate the transmitting efficiency of the MRC coils with the ICN

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

where the parameters S_{11}, S_{22} are reflection coefficients; S_{11}, S_{11} are transmission coefficients; a_1, b_1 are the traveling wave variables at port 1; and a_2, b_2 are the traveling wave variables at port 2. The network analyzer was used to measure S_{21} and S_{11}. The transmitting efficiency was

$$\eta = \frac{S_{21}^2}{1 - S_{11}^2}. \tag{21}$$

Fig. 15(a) shows the transmitting efficiency of the MRC coils with the RCN and ICN when the distance between the coils varies. Fig. 15(b) shows the transmitting efficiency with the RCN and ICN when we have the horizontal misalignment between the coils. The efficiency with the ICN was almost close to the value with the RCN even though we connected two more passive components in the ICN.

V. Simulation Results of a Class Φ_2 Inverter Connected to an ICN

To demonstrate the performance of the ICN with a resonant inverter, we designed and simulated a class Φ_2 inverter connected to an ICN in LTSPICE. First, the inverter provided

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

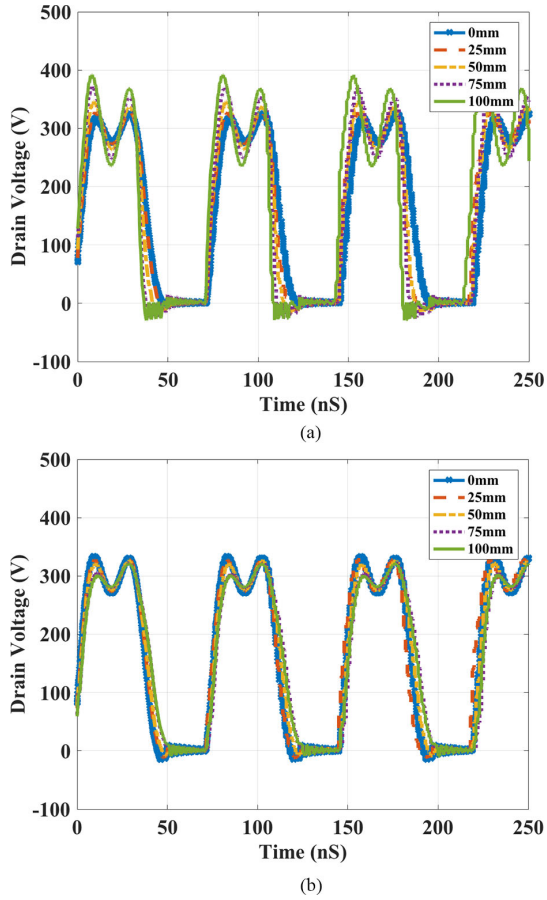


Fig. 16. Simulated drain voltages in the class Φ_2 inverter with MRC coil impedance. (a) Without the ICN. (b) With the ICN.

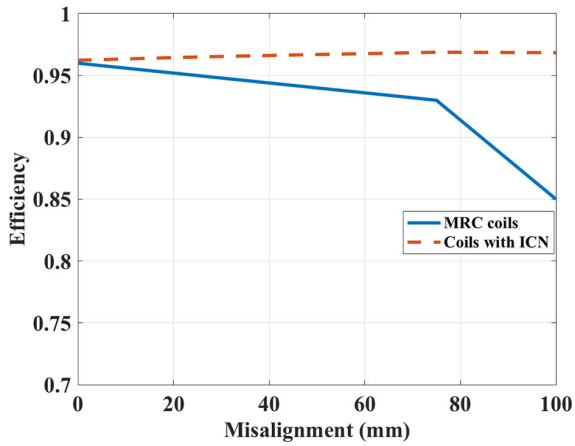


Fig. 17. Comparison of the inverter efficiency when the horizontal alignment between coils varies in simulation. The solid line shows the efficiency without the ICN and the dash line shows the efficiency with the ICN.

600 W output power to 16 Ω, the reference impedance of the MRC coils when the input voltage was 150 V and the switching frequency was 13.56 MHz.

First, we simulated the class Φ_2 inverter without the ICN when the load values vary due to the horizontal misalignment between MRC coils. Fig. 16(a) shows the drain voltages of the class Φ_2 inverter without the ICN. While the maximum drain voltage without horizontal misalignment was 320 V, the maximum drain voltage with 100 mm misalignment reached 400 V. In addition, when the horizontal misalignment was 100 mm, the inverter efficiency dropped to 85%, as shown in Fig. 17. In contrast, connecting the ICN to the inverter allowed us to maintain ZVS operation and constant maximum voltage (300 V) when horizontal misalignments occurred. Fig. 16(b) shows the drain voltages of the class Φ_2 inverter with the ICN when horizontal alignments vary. Also, the inverter achieved 95% efficiency, as shown in Fig. 17.

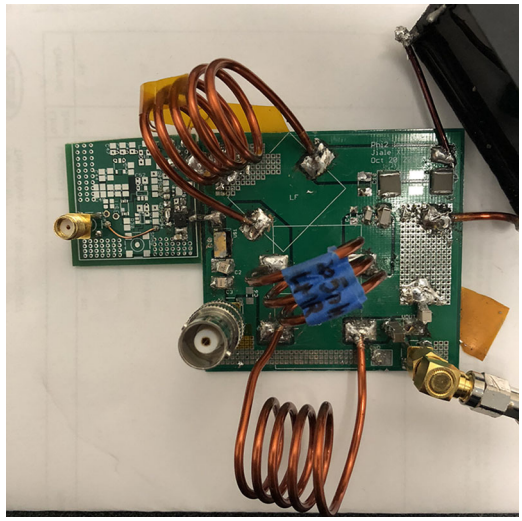


Fig. 18. Assembled class Φ_2 inverter.

TABLE III
PARAMETERS OF THE CLASS Φ_2 INVERTER

Parameter	Value	Units
L_f	300	nH
L_{MR}	205	nH
L_s	290	nH
C_{MR}	170	pF
C_p	342	pF
C_s	2.16	nF
GAN	GS66508	

VI. EXPERIMENTAL RESULTS OF A CLASS Φ_2 INVERTER CONNECTED TO AN ICN

We implemented a class Φ_2 inverter to evaluate the ICN performance with MRC coils as shown in Fig. 18. Solenoidal-type inductors were used and a GS66508T eGaN FET was selected to operate at 13.56 MHz. An LM5114 gate driver provided gate voltage to the switching device. Table III shows the implemented parameters of the class Φ_2 inverter. This inverter was designed to provide 350 W with a 150 V input voltage and 600 W with a 200 V input voltage. We connected the inverter to the designed ICN and applied input voltage, 150 and 200 V. In our experiments, we retuned the MRC coils to have the same impedance

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

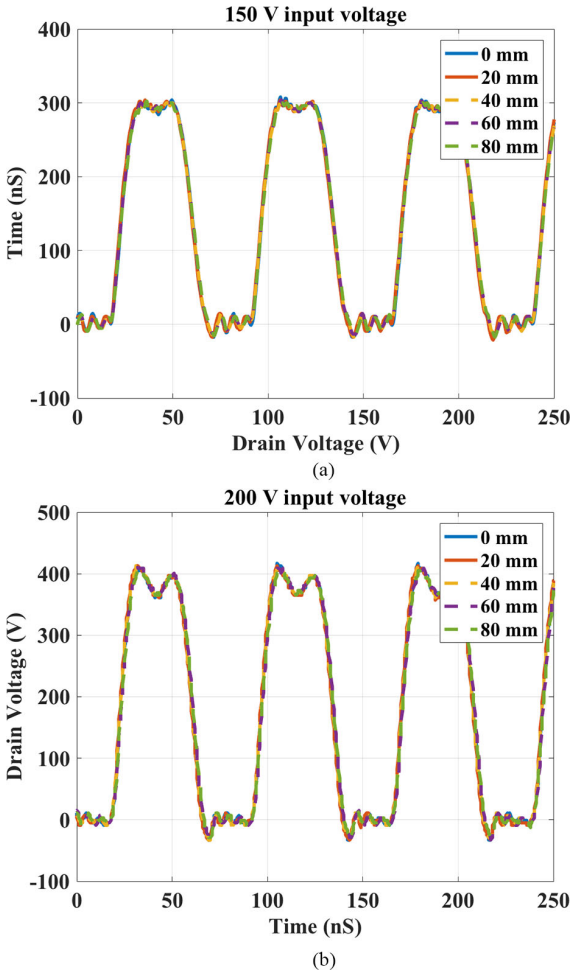


Fig. 19. Measured drain voltages in the class Φ_2 inverter with ICN and MRC coils. (a) 150 V input voltage. (b) 200 V input voltage.

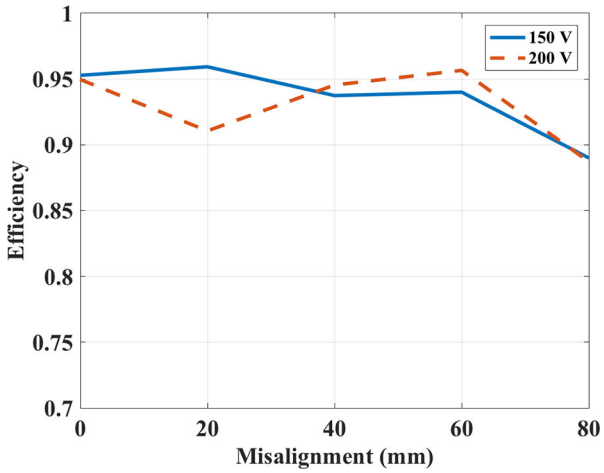


Fig. 20. Measured inverter efficiency with ICN.

value as each other. Therefore, Z_o was 10 Ω, L_{RCN} was 117 nH, and C_{RCN} was 1.17 nF. Also, we added a 700 pF capacitor only as a PCN because the MRC coil impedance with the RCN was inductive and easily aligned with the real axis with a capacitor only.

Fig. 19(a) and (b) shows the measured drain voltage waveforms when the horizontal misalignment between the transmitting and receiving coils varies from 0 to 80 mm. In our experiments, we applied 150 and 200 V of input voltage to the inverter. As shown in this figure, the drain voltage waveforms are identical to each other and maintain ZVS and zero dv/dt operation.

Fig. 20 shows the efficiency of the inverter with misalignments. At 150 V input voltage, the inverter efficiency was 95% (no misalignment) to 88% (80 mm misalignment). At 200 V input voltage, the inverter efficiency was 95% (no misalignment) to 86% (80 mm misalignment). Because of losing ZVS in the inverter, we did not measure the efficiency of the inverter and MRC coils without the ICN when there are distance variations or misalignments.

VII. Conclusion

This paper presents the design and implementation of an ICN to mitigate distance and horizontal alignment variations in a WPT system. While an ICN compresses the magnitude variation of an impedance through an RCN, it also reduces the variation in a phase of impedance through a PCN, when the distance or horizontal alignment between MRC coils varies. Therefore, an ICN helps us to obtain high efficiency while operating with a resonant inverter, which is sensitive to load variations.

We designed and implemented the MRC coils to deliver power over a 150 mm distance at 13.56 MHz. The coil impedance was measured by the network analyzer for designing the ICN. Adding the ICN allows us to compress the magnitude of the coil impedance to $0.74 * Z_{\mathrm{center}} < Z_{\mathrm{IN}} < 1.8 * Z_{\mathrm{center}}$ with distance variations. Also, when the alignment between coils varies, the magnitude variation is only 0.794 times the reference impedance value. In order to address the phase shift in the MRC coils with the RCN, the PCN was connected to them. By adding an inductor and capacitor to the MRC coils with the RCN, the phase of the impedance reduced it to nearly zero when the distance and alignment varied. Finally, we designed a single-ended class Φ_2 inverter and connected it to the designed ICN with MRC coils in simulations and experiments. The ICN allows us to maintain ZVS and constant maximum drain voltage while the alignment between coils varies. In conclusion, using an ICN successfully compress impedance variations in MRC coils with the deviations of the coil positions. As a result, the WPT system enables us to improve a WPT system performance in high-power applications, such as AGVs.

Appendix

As we described in Section II-B, when there are misalignments between transmitting and receiving coils, the impedance variations are

$$Z_{\mathrm{Load}_i} = R_{L,i} - jX_{L,i}, 1 < i < n. \tag{22}$$

As the misalignment increases, the absolute value $|X_{L,n}|$ increases. To design the PCN, we need to add an inductor, which has $|X_{L,n}|$ value to compensate the largest variation. This inductance value helps other absolute reactance values, $|X_{L,n}|(1 < i < n-1)$, decrease, but these values now have inductive values. Thus, we have the impedance values as

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.

follows:

$$Z_{\mathrm{PCN},L,i} = R_{\mathrm{PCN},i} - jX_{\mathrm{PCN},i} = R_{L,i} - j(X_{L,i} - X_{L,n}), 1 < i < n \quad (23)$$

where $X_{L,n} = wL_{\mathrm{PCN}}$. Because $|X_{L,n}|$ is bigger than other reactance values, $Z_{\mathrm{PCN},i}$ has inductive term. To compensate for these inductive values, we need to add a capacitor (C_{PCN}) in parallel. To find the capacitance value, we calculate the admittance values of $Z_{\mathrm{PCN},i}$, as shown below

$$Y_{\mathrm{PCN},L,i} = \frac{1}{Z_{\mathrm{PCN},L,i}} = \frac{1}{R_{L,i} - j(X_{L,i} - X_{L,n})} = G_{\mathrm{PCN},i} - jB_{\mathrm{PCN},i}, 1 < i < n. \quad (24)$$

Similarly, when we calculate C_{RCN}, As the same as in the previous case, selecting L_{PCN}, we need to add the largest reactance value $|B_{\mathrm{PCN},n}|$ to this network. Then, we obtain new impedance values

$$Z_{\mathrm{PCN},C,i} = \frac{1}{Z_{\mathrm{PCN},L,i}} = \frac{1}{G_{\mathrm{PCN},i} - j(B_{\mathrm{PCN},i} - B_{\mathrm{PCN},n})} = G_{\mathrm{PCN},i} - jB_{\mathrm{PCN},i}, 1 < i < n, \quad (25)$$

where $B_{L,n} = wC_{\mathrm{PCN}}$. This process, selecting one inductor and one capacitor, is one cycle in designing the PCN. If we still have some reactance values, which indicate phase shifts, then this process can be repeated until we have very small or zero phase shifts.

REFERENCES

[1] J. Dai and D. C. Ludois, "Capacitive power transfer through a conformal bumper for electric vehicle charging," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 4, no. 3, pp. 1015–1025, Sep. 2016.

[2] T. Mishima and E. Morita, "High-frequency bridgeless rectifier-based ZVS multi-resonant converter for inductive power transfer featuring high-voltage GaN-HFET," *IEEE Trans. Ind. Electron.*, vol. 64, no. 11, pp. 9155–9164, Nov. 2017.

[3] G. Ewing, "High-efficiency radio-frequency power amplifiers," Ph.D. dissertation, Dept. Elect. Eng., Oregon State Univ., Corvallis, OR, USA, 1964.

[4] M. K. Kazimierczuk and J. Jozwik, "Resonant dc/dc converter with class-e inverter and class-e rectifier," *IEEE Trans. Ind. Electron.*, vol. 36, no. 4, pp. 468–478, Nov. 1989.

[5] J. M. Rivas, Y. Han, O. Leitermann, A. Sagneri, and D. J. Perreault, "A high-frequency resonant inverter topology with low voltage stress," in *Proc. IEEE Power Electron. Spec. Conf.*, 2007, pp. 2705–2717.

[6] A. Sepahvand, A. Kumar, K. Afridi, and D. Maksimović, "High power transfer density and high efficiency 100 MHz capacitive wireless power transfer system," in *Proc. IEEE 16th Workshop Control Model. Power Electron.*, Jul. 2015, pp. 1–4.

[7] S. Aldhaher, D. C. Yates, and P. D. Mitcheson, "Design and development of a class ef_2 inverter and rectifier for multimegahertz wireless power transfer systems," *IEEE Trans. Power Electron.*, vol. 31, no. 12, pp. 8138–8150, Dec. 2016.

[8] J. Choi, D. Tsukiyama, Y. Tsuruda, and J. M. Rivas, "High-frequency, high-power resonant inverter with eGaN FET for wireless power transfer," *IEEE Trans. Power Electron.*, vol. 33, no. 3, pp. 1890–1896, Mar. 2018.

[9] F. Lu, H. Zhang, and C. Mi, "A two-plate capacitive wireless power transfer system for electric vehicle charging applications," *IEEE Trans. Power Electron.*, vol. 33, no. 2, pp. 964–969, Feb. 2018.

[10] J. Choi, D. Tsukiyama, and J. Rivas, "Evaluation of a 900 V SiC MOSFET in a 13.56 MHz 2 kW resonant inverter for wireless power transfer," in *Proc. IEEE 17th Workshop Control Model. Power Electron.*, Jun. 2016, pp. 1–6.

[11] A. Kurs, A. Karalis, R. Moffatt, J. D. Joannopoulos, P. Fisher, and M. Soljačić, "Wireless power transfer via strongly coupled magnetic resonances," *Science*, vol. 317, no. 5834, pp. 83–86, 2007.

[12] S. Y. R. Hui, W. Zhong, and C. K. Lee, "A critical review of recent progress in mid-range wireless power transfer," *IEEE Trans. Power Electron.*, vol. 29, no. 9, pp. 4500–4511, Sep. 2014.

[13] F. Lu, H. Zhang, H. Hofmann, and C. Mi, "A high efficiency 3.3 kW loosely-coupled wireless power transfer system without magnetic material," in *Proc. IEEE Energy Convers. Congr. Expo.*, Sep. 2015, pp. 2282–2286.

[14] N. K. Trung and K. Akatsu, "Design high power and high efficiency inverter operating at 13.56 MHz for wireless power transfer systems," in *Proc. IEEE Energy Convers. Congr. Expo.*, Sep. 2016, pp. 1–8.

[15] D. H. Tran, V. B. Vu, and W. Choi, "Design of a high efficiency wireless power transfer system with intermidiate coils for the on-board chargers of electric vehicles," *IEEE Trans. Power Electron.*, vol. 33, no. 1, pp. 175–187, Jan. 2018.

[16] C. Deng, G. Zhu, and R. D. Lorenz, "Mhz frequencies, kW, 30 cm gap wireless power transfer with low air gap flux density and high efficiency using surface spiral winding coils," in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, Mar. 2017, pp. 1606–1613.

[17] Y. Lim, H. Tang, S. Lim, and J. Park, "An adaptive impedance-matching network based on a novel capacitor matrix for wireless power transfer," *IEEE Trans. Power Electron.*, vol. 29, no. 8, pp. 4403–4413, Aug. 2014.

[18] B. L. Cannon, J. F. Hoburg, D. D. Stancil, and S. C. Goldstein, "Magnetic resonant coupling as a potential means for wireless power transfer to multiple small receivers," *IEEE Trans. Power Electron.*, vol. 24, no. 7, pp. 1819–1825, Jul. 2009.

[19] D. Patil, M. Sirico, L. Gu, and B. Fahimi, "Maximum efficiency tracking in wireless power transfer for battery charger: Phase shift and frequency control," in *Proc. IEEE Energy Convers. Congr. Expo.*, Sep. 2016, pp. 1–8.

[20] Y. Han, O. Leitermann, D. Jackson, J. Rivas, and D. Perreault, "Resistance compression networks for radio-frequency power conversion," *IEEE Trans. Power Electron.*, vol. 22, no. 1, pp. 41–53, Jan. 2007.

[21] J. Xu and D. S. Ricketts, "An efficient, watt-level microwave rectifier using an impedance compression network (ICN) with applications in outphasing energy recovery systems," *IEEE Microw. Wireless Compon. Lett.*, vol. 23, no. 10, pp. 542–544, Oct. 2013.

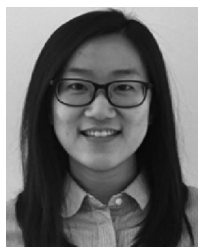
[22] J. Choi and J. Rivas, "Implementing an impedance compression network to correct misalignment in a wireless power transfer system," in *Proc. IEEE 18th Workshop Control Model. Power Electron.*, Jul. 2017, pp. 1–8.

[23] M. Kamon, M. J. Tsuk, and J. K. White, "Fasthenry: A multipole-accelerated 3-d inductance extraction program," *IEEE Trans. Microw. Theory Techn.*, vol. 42, no. 9, pp. 1750–1758, Sep. 1994.



Jungwon Choi (S'13) received the B.Eng. degree from Korea University, Seoul, South Korea, and the M.S.E. degree from the University of Michigan, Ann Arbor, MI, USA, in 2009 and 2013, respectively. She is currently working toward the Ph.D. degree at the Stanford University Power Electronics Research Laboratory, Stanford University, Stanford, CA, USA.

Her research interests include high-frequency and high-power converters with WBG devices for wireless power transfer applications.



Jiale Xu (S'17) received the B.A.Sc. degree in electrical engineering from the University of Toronto, Toronto, ON, Canada, in 2015, and the M.S. degree in electrical engineering from Stanford University, Stanford, CA, USA, in 2017. She is currently working toward the Ph.D. degree in electrical engineering at Stanford University.

Her research interests include resonant converters, high-frequency power converters, and wide bandgap power devices.

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.



Rawad Makhoul (S'16) received the Engineer's degree from the National Institute of Applied Sciences of Lyon (INSA Lyon), Villeurbanne, France, and the M.S. degree in electronic engineering from the Politecnico di Torino, Turin, Italy, in 2016. He is currently working toward the Ph.D. degree at the French Alternative Energies and Atomic Energy Commission (CEA), Grenoble, France.

His research interests include very high frequency converters and self-oscillating converter topologies.



Juan M. Rivas Davila (S'01–M'06) was born in Mexico, Mexico. He received the B.A.Sc. degree from the Monterrey Institute of Technology, Monterrey, Mexico, in 1998, and the S.M. and Sc.D. degrees from the Laboratory of Electromagnetic and Electronic Systems, Massachusetts Institute of Technology, Cambridge, MA, USA, in 2003 and 2006, respectively.

From 2007 to 2011, he was a Power Electronics Engineer with the High-Frequency Power Electronics Group, General Electric Global Research Center, Niskayuna, NY, USA. From 2011 to 2013, he was an Assistant Professor with the University of Michigan, Ann Arbor, MI, USA. In 2014, he joined the Electrical Engineering Department, Stanford University, Stanford, CA, USA, as an Assistant Professor. His current research interests include power electronics, RF power amplifiers, resonant converters, soft-switching topologies, and the design of air-core passive components for VHF power conversion.

Authorized licensed use limited to: UCLA Library. Downloaded on June 05,2025 at 06:32:50 UTC from IEEE Xplore. Restrictions apply.